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Math 400

Exam 1 Study Guide

Exam 1 will be made available on Thursday, September 17 10:00 am and will be due by Monday, September 21 11:59 pm.

Exam 1 will be based on Sections 2.1, 2.2, 2.3, 2.4, 2.5, 2.6

In order to prepare for your exam, you should review the Web Assign Homework, Non-Online Homework, and the additional problems that I provided below.

The Exam will be made up of parts on **WebAssign** and a part on **Canvas**.

WebAssign: The parts of the exam on **WebAssign will be timed**. Once you start a part of the exam, the timer will start. You must finish that part of the exam in one sitting. You cannot exit that part of the exam and come back to work on it later.

You do not have to show your work for the problems done on WebAssign. There will be no partial credit for the work done on WebAssign. You will only be given **one attempt to submit** your answers in WebAssign.

Canvas: This part of the exam will not be timed. You will have to download this part of the exam from Canvas and write your solutions on paper. You will then need to take pictures of your solutions and convert it to a single PDF. After that, you will need to upload these problems to Canvas by the due date (Monday, September 21 11:59 pm).

You will need to **show an appropriate amount of work** for each problem on the part of the exam downloaded from Canvas. If you are unsure of what I mean by "**appropriate amount of work**", then you should watch my lecture videos and look at solutions to the practice problems that I include with this study guide.

If you've been keeping up with the homework and studied for the exam, then Exam 1 shouldn't take more than 2 to 3 hours to complete.

**This is an exam and so you shouldn't be working on the exam with other people and/or using apps/websites that can solve the problems for you.

The following questions will be answered when I post the instructions to Exam 1 on Thursday, September 17.

How many problems will Exam 1 have?

How much time will I be given to complete the parts of the exam on WebAssign?

How many parts of the exam will be put on WebAssign?

How many points will be on the Exam? How many points will the WebAssign parts of the Exam be worth and how many points will the Canvas part of the exam be worth?

How and where do I submit the Canvas part of the exam?

Here are some practice problems to work on. This is not a replacement for doing the homework and this doesn't cover everything from the homework.

Chapter 2 Review

Concept Check page 165 # 2, 4, 5, 6, 7, 8, 9

True or False Quiz page 166 # 7, 8, 9, 10, 11, 12, 13, 14, 15, 16

Exercises pages 166 – 167 # 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 23, 24, 25, 29, 30, 31, 32, 33, 34

Concept: Find limits graphically.

Chapter 2 Review Exercises page 166 # 1

1. Let $f(x)=\begin{cases}\sqrt[3]{x-1}, \text{ if } x \geq 0 \\ -\frac{3}{4}x-1, \text{ if } x<0\end{cases}$.

(a.) Graph the piecewise function. Find the domain and range of the function.

(b.) Use the graph of f to find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

(c.) Find $\lim_{x\to 0} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

(d.) Find $\lim_{x\to 8} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

2. Let $f(x)=\begin{cases}\log_2(-x)\text{ , if } x<0\\ \dfrac{1}{x^3}\text{ , if } x>0\end{cases}$.

(a.) Graph the piecewise function. Find the domain and range of the function.

(b.) Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

(c.) Find $\lim_{x\to 0} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

3. Let $f(x)=\begin{cases}\sqrt{x}+2,\ \text{if } x\geq 0\\ 2,\text{ if } -3<x<0\\ \dfrac{4}{3}x+8,\text{ if } x\leq -3\end{cases}$.

(a.) Graph the function f. Find the domain and range of the function.

(b.) Use the graph of f to find $\lim_{x\to 0^-} f(x)$.

(c.) Use the graph of f to find $\lim_{x\to 0^+} f(x)$.

(d.) Find $\lim_{x\to 0} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

(e.) Use the graph of f to find $\lim_{x\to 3^-} f(x)$.

(f.) Use the graph of f to find $\lim_{x\to 3^+} f(x)$.

(g.) Find $\lim_{x\to 3} f(x)$, if it exists. If it does not exist, then explain why using appropriate limit notation.

3. (continued)

(h.) Use the graph of f to find $\lim_{x\to 4^-} f(x)$.

(i.) Use the graph of f to find $\lim_{x\to 4^+} f(x)$.

(j.) Find $\lim_{x\to 4} f(x)$, if it exist. If it does not exist, then explain why using appropriate limit notation.

Concept: Find limits at finite numbers algebraically. These limits have the indeterminate form 0/0.

Chapter 2 Review Exercises pages 167 # 5, 7, 8, 10, 11, 12

4. Find the following limits algebraically.

$(a.)\ \lim_{x\to -1} \dfrac{x^4+4x^2-5}{x^3+1}$

$(b.)\ \lim_{x\to 7/5} \dfrac{5x^2+3x-14}{15x^2-16x-7}$

$(c.)\ \lim_{h\to 0} \dfrac{\sqrt{12+h}-2\sqrt{3}}{h}$

$(d.)\ \lim_{x\to 625} \dfrac{x-625}{25-\sqrt{x}}$

$(e.)\ \lim_{h\to 0} \dfrac{\dfrac{1}{(9+h)^2}-\dfrac{1}{9^2}}{h}$

$(f.)\ \lim_{x\to 2} \dfrac{x^4-16}{2-x}$

4. (continued)

$(g.)\ \lim_{x\to a}\dfrac{x^2-a^2+(x-a)^{100}}{x-a}$

$(h.)\ \lim_{x\to 1}\left(\dfrac{x^2}{x^2-5x+4}-\dfrac{1}{x^2-5x+4}\right)$

5. Find the following limits algebraically.

$(a.)\ \lim_{x\to\pi}\dfrac{\cos^2 x+3\cos x+2}{\cos x+1}$

$(b.)\ \lim_{x\to 0}\dfrac{\dfrac{\mathbf{1}}{\mathbf{2+\sin x}}-\dfrac{\mathbf{1}}{\mathbf{2}}}{\sin x}$

6. Find the following limit, if it exists. If it does not exist, then explain why using appropriate limit notation.

$\lim_{x\to 4}\dfrac{x^2-16}{|x-4|}$

Concept: Find limits using the limit laws and continuity.

Chapter 2 Review Exercises page 167 # 3

7. Find the following limits.

$(a.)\ \lim_{x\to 0}(4x^3-2x^2+10)$

$(b.)\ \lim_{x\to 4}\dfrac{x-10}{x^3+2x}$

$(c.)\ \lim_{x\to 3}\sqrt{\dfrac{x^2+9x}{x-1}}$

$(d.)\ \lim_{x\to\pi}2e^{x/\pi}\cos x$

Concept: Use the Squeeze theorem to find a limit.

Chapter 2 Review Exercises page 167 # 23, 24

8. Use the squeeze theorem to verify that $\lim_{x\to 0} x^4 \sin\left(\frac{1}{x^3}\right) = 0$.

Use the definition of continuity to show that a function is continuous or discontinuous at $x = a$. Find the intervals where a function is continuous.

Chapter 2 Review Exercises page 167 # 29, 30, 31, 32

9. Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), \text{if } x \neq 0 \\ 0, \text{if } x = 0 \end{cases}$. Is f continuous at $x = 0$? Explain.

10. Let $f(x) = \dfrac{|x-3|}{x-3}$.

(a.) Sketch a graph of *f*.

(b.) Find all values of x, where *f* is discontinuous. Use the definition of continuity to explain your answer.

11. Find a value of *k* that will make the following function continuous.

$$f(x) = \begin{cases} x^2 + 3 \text{ , if } x < 0 \\ -4x + k, \text{ if } x \geq 0 \end{cases}$$

12. Determine whether the following function is continuous at a. Use the definition of continuity to verify your answer.

$$f(x) = \begin{cases} \dfrac{x^2 + x}{x+1} \text{ , if } x \neq -1 \\ x^2 + 3x, \text{ if } x = -1 \end{cases}, \quad a = -1$$

13. How would you define $f(2)$ in order to make f continuous at 2? Verify that this satisfies the definition of continuity.

$$f(x)=\frac{x^2-4}{x^2+x-6}$$

14. Determine the values of *x* at which the following function *f* has discontinuities.

$$f(x)=\frac{1+\sin x}{\cos x}$$

15. Sketch the graph of the function. Use the definition of continuity to explain why the function is discontinuous at $x=\frac{\pi}{4}$.

$$f(x)=\begin{cases}\cos x, \text{ if } x\geq\frac{\pi}{4}\\ e^x, \text{ if } x<\frac{\pi}{4}\end{cases}$$

Concept: Find limits at infinity algebraically. These limits have the indeterminate form $\frac{\infty}{\infty}$ or $\infty-\infty$.

Chapter 2 Review Exercises page 167 # 13, 14, 16, 17, 20

16. Find the following limits algebraically.

(a.) $\lim\limits_{x\to\infty}\dfrac{17x^4-9x+1}{5x^4+9x^2+10x}$

(b.) $\lim\limits_{x\to-\infty}\dfrac{-4x^5+3x^2-8x+1}{1-16x^2}$

17. Find the limit algebraically.

(a.) $\lim\limits_{x\to\infty}\dfrac{3\cdot 2^{3x}+2^x}{2^{7x}-11\cdot 2^x}$

(b.) $\lim\limits_{x\to\infty}\dfrac{e^{4x}+e^{-4x}}{e^{4x}-e^{-4x}}$

18. Determine the end behavior of the following function. You will need to find $\lim_{x\to\infty} h(x)$ and $\lim_{x\to-\infty} h(x)$ algebraically. Indicate the equation(s) of any horizontal asymptote(s).

$$h(x)=\frac{\sqrt{3x^{18}+x^{10}+4}}{-7x^{9}+4x^{2}-9x}$$

19. Find the limit algebraically.

$$\lim_{x\to\infty}\left(\sqrt{3x^{2}+8x+6}-\sqrt{3x^{2}+3x+1}\right)$$

Concept: Find limits at infinity analytically.

Chapter 2 Review Exercises page 167 # 9, 15, 18, 19

20. Find the following limits analytically.

(a.) $\lim_{x\to-\infty} \arctan\left(e^{|2x-1|}\right)$

(b.) $\lim_{x\to\infty} \sin\left[\operatorname{arcsec}\left(\frac{x-4}{\sqrt{x}-2}\right)\right]$

(c.) $\lim_{x\to\infty} \ln\left(\ln\left(\ln x\right)\right)$

Concept: Find limits at a finite number analytically.

21. Find the following limits analytically. If the limit is ∞ or $-\infty$, then state this.

$(a.)$ $\lim_{x\to 1^-} \ln(\cos^{-1} x)$

$(b.)$ $\lim_{x\to 8^-} \dfrac{x+8}{(x-8)^2(2x+3)}$

$(c.)$ $\lim_{x\to 2} \arcsin\left(\dfrac{2-x}{4\left(\sqrt{2}-\sqrt{x}\right)}\right)$

$(d.)$ $\lim_{x\to \pi^+} \dfrac{1}{x^2}\csc(x-\pi)$

Concept: Intermediate Value Theorem

Chapter 2 Review Exercises page 167 # 33, 34

Concept: Use the Precise Definition of a limit to prove a limit.

22. Prove the statement by using the ε, δ definition of a limit.

$\lim_{x\to 2}(-14x+10) = -18$

23. Prove the statement by using the ε, δ definition of a limit.

$\lim_{x\to -\frac{1}{3}} \dfrac{9x^2-1}{3x+1} = -2$

24. Use the precise definition of an infinite limit to prove the following.

$$\lim_{x\to 9} \frac{1}{(x-9)^2} = \infty$$